

SHARORAH, Saudi Arabia

WMO: 411360

Lat: 17.467N

Lon: 47.121E

Elev: 720

StdP: 92.97

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	10.7	-10.1	1.7	21.2	-7.9	2.1	23.1	9.4	19.9	8.4	19.5	1.3	80	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	43.1	21.4	42.7	21.3	42.0	21.2	27.3	39.6	25.3	38.0	23.8	37.8	3.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.6	36.7	21.2	17.3	35.3	19.1	15.2	33.5	92.1	39.0	82.3	38.0	75.7	37.8	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.8	6.9	5.1	44.5	2.3	0.7	3.5	45.0	2.1	45.4	0.9	45.8	-0.8	46.4		
(5)			0.4	25.8	2.3	2.5	-1.2	27.6	-2.6	29.0	-3.8	30.4	-5.5	32.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	19.2	22.4	26.2	29.9	32.9	34.3	35.4	35.1	32.6	27.8	23.5	19.8
	DBStd	6.21	3.43	3.52	2.95	2.34	1.56	1.47	1.23	1.44	1.86	2.18	2.43	3.25
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	63	32	8	0	0	0	0	0	0	0	0	1	22
	CDD10.0	6680	286	348	503	598	710	730	786	777	679	552	406	305
	CDD18.3	3701	59	123	245	348	452	480	528	519	429	293	156	69
	CDH23.3	59233	713	1504	3192	5236	7618	8505	9452	9153	7196	4147	1769	749
CDH26.7	39212	241	703	1741	3267	5234	6148	6985	6686	4923	2346	712	227	
Wind	WSAvg	3.2	3.2	3.5	3.8	3.7	3.4	2.9	3.1	3.4	3.0	2.8	3.1	3.0
Precipitation	PrecAvg	63	1	3	15	16	5	5	5	6	3	2	1	0
	PrecMax	266	10	34	103	112	25	88	50	65	81	28	10	1
	PrecMin	3	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	59	3	7	29	29	8	17	11	13	15	6	2	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	36.2	38.8	41.2	42.2	43.9	44.1	43.5	42.2	38.2	34.1	33.0
		MCWB	16.9	17.5	18.5	21.1	19.9	21.5	21.7	21.9	21.1	19.6	19.2	18.1
	2%	DB	31.1	34.3	37.1	39.2	41.4	43.0	43.1	42.9	41.2	37.1	32.9	31.0
		MCWB	16.9	17.4	18.6	20.1	20.3	20.9	21.4	22.0	20.9	19.1	18.0	17.5
	5%	DB	29.2	33.0	35.9	38.2	40.8	42.1	42.8	42.1	40.6	36.0	31.9	29.1
		MCWB	15.9	17.6	18.3	19.5	20.4	20.4	21.4	22.0	20.6	18.5	17.6	16.2
	10%	DB	27.7	31.1	34.2	37.2	39.9	41.2	42.0	41.7	39.8	34.9	30.8	27.4
		MCWB	15.3	16.6	18.1	19.1	20.4	20.1	21.3	21.9	20.3	18.3	17.0	15.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	21.2	22.4	23.7	25.8	26.8	29.9	29.5	27.0	23.4	21.6	20.2
		MCDB	30.6	32.4	33.2	36.7	38.3	40.1	41.2	40.1	39.1	33.0	31.3	29.9
	2%	WB	18.2	19.5	21.0	22.5	24.3	24.2	27.2	27.5	23.4	21.6	19.7	18.4
		MCDB	28.9	31.5	32.4	35.8	38.0	37.1	39.7	39.6	37.8	32.3	29.7	29.1
	5%	WB	16.9	18.3	20.0	21.6	23.1	22.8	24.5	25.5	21.9	20.0	18.7	17.1
		MCDB	27.6	30.7	31.6	35.4	37.4	38.8	38.3	38.5	38.4	33.9	29.7	27.6
	10%	WB	15.8	17.1	19.1	20.7	21.9	21.7	22.9	23.7	20.9	19.0	17.7	16.1
		MCDB	26.0	29.5	31.5	34.9	37.1	39.7	38.7	37.9	38.2	33.5	29.0	26.2
Mean Daily Temperature Range	5% DB	MDBR	14.2	14.8	14.6	14.1	14.0	14.7	14.0	13.5	14.2	14.1	13.7	13.8
		MCDBR	16.6	17.2	16.4	15.3	15.1	15.3	14.6	13.9	14.6	14.6	14.3	15.0
		MCWBR	8.3	8.1	7.4	6.8	6.9	6.7	6.3	6.1	6.5	6.7	7.1	7.2
	5% WB	MCDBR	15.3	15.6	14.3	13.5	13.4	14.1	13.3	13.1	14.3	13.7	13.5	14.1
		MCWBR	8.4	8.3	7.5	6.9	7.7	7.7	7.9	7.5	7.4	7.1	7.3	7.6
Clear-Sky Solar Irradiance	taub		0.429	0.465	0.523	0.527	0.583	0.672	0.715	0.677	0.561	0.438	0.410	0.405
	taud		1.941	1.839	1.720	1.746	1.615	1.444	1.428	1.494	1.662	1.925	2.009	2.005
	Ebn at Noon		844	834	800	798	744	674	647	677	757	848	857	854
	Edh at Noon		178	207	240	236	265	312	317	299	251	187	164	162
All-Sky Solar Radiation	RadAvg		5.41	6.16	6.85	7.30	7.55	7.44	7.09	7.02	6.99	6.63	5.75	5.27
	RadStd		0.22	0.20	0.23	0.26	0.20	0.24	0.24	0.18	0.08	0.15	0.13	0.26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	+80
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page